

*Department of Energy and Resources,
Government of Prjmnestwe*

NATIONAL ENERGY POLICY 2025

Drafted by:

Charlie Barrett-Lennard
Clover Zijing Luo
Haritha Vattamparambil Ajithkumar
Mufahim Abqary

EXECUTIVE SUMMARY

Prjmnestwe's energy is at a critical juncture, with a considerable but incomplete shift from a historically fossil fuel-dominated economy to rising renewable electricity production. Decarbonisation efforts have been successful in the electricity sector, with **renewables now accounting for 61.8% of electricity in 2020**. However, the country faces ongoing challenges in the heating, industry and transport sectors, which continue to rely heavily on non-renewables.

This National Energy Policy (NEP) 2025 outlines a framework to accelerate the transition towards a sustainable and low-carbon energy future for Prjmnestwe. Given the limited funds, resources, and expected climate change implications, the policy **targets high payback, high impact, cost-efficient strategies that can be easily implemented** to ensure maximum benefit with minimal outlay.

The policy prioritises two key areas for immediate and comprehensive action:

Part 1- Aims to reshape national mobility by promoting the **adoption of electric vehicles (EVs)** and developing a modern, **efficient public transport system**. Drawing inspiration from the successful model of Norway's comprehensive EV incentives and strategic public transport network design principles, to significantly reduce reliance on fossil fuels in the transport sector.

Part 2- Targets a substantial **reduction in energy demand from residential heating**. Leveraging proven strategies like mandatory energy performance standards for buildings, as exemplified by Germany's Building Energy Act (GEG), and supported by financial incentives and public engagement initiatives.

Prjmnestwe will drive the adoption of policies through a **phased approach of incentives, infrastructure development, and public awareness campaigns**

Table of Contents

EXECUTIVE SUMMARY.....	2
INTRODUCTION.....	4
ENERGY PROFILE.....	5
POLICY MEASURES AND STRATEGIES.....	6
Part 1 – Reshaping National Mobility.....	6
Policy 1 – Electric Vehicles.....	6
Policy 2 – Public Transport.....	7
Part 2- Energy Efficient Household Heating.....	10
BACKGROUND.....	10
ENERGY MIX FOR DOMESTIC HEATING.....	10
Evidence for Energy Efficiency measures- Germany’s Building Energy Act (GEG).....	10
DECARBONISATION TARGETS.....	11
PROPOSALS.....	12
Section 1- Establishment of Energy Efficiency Standards.....	12
Section 2- Transition to Electricity-based Heating.....	14
Section 3- Public Awareness and Behavioural Change.....	15
Policy Implementation Framework.....	15
Performance Indicator Metrics.....	15
References.....	17
Acknowledgements.....	17

List of Figures

Figure 1. Multi-scaler, multimodal, spoke-hub public transit network diagram9

Figure 2. Heat losses and gains in the absence of effective insulation and airtightness measures.....13

Figure 3. Rating System Mandates for New and Existing Constructions.....14

List of Tables

Table 1. Energy Mix as per National Data indicator of Emission Profile.....6

Table 2. Development timeline for rural rail network10

Table 3. Energy mix for domestic heating.....11

Table 4. Streamlined Measures for Heating Demand Management12

Table 5. Minimum requirement for EPC accreditation aligned with global recommendations.....14

Table 6. Gantt Chart for NEP 2025, Implementation timelines and expected achievement.....16

INTRODUCTION

The Government of Prjmnestwe is committed to providing a **secure, affordable, and sustainable energy future**. The energy landscape is currently undergoing a dramatic transition. Historically reliant on fossil fuels, the country has achieved progress in the power generation industry, with renewables now accounting for 61.8% of the electricity mix in 2020 illustrating Prjmnestwe's readiness to embrace a net-zero future.

The transport sector accounts for a considerable **25% of total energy consumption** and the **most amount of carbon emissions in 2019**, with petrol and diesel fuels overwhelmingly powering cars, vans, and a growing freight transport sector. However, the world contribution of carbon emissions by transportation is only 13% in 2019 (Dimancheve, 2022). Public transport, such as trains and buses, and motorcycles are not widely utilised. Since **private cars are the dominant mode of transportation, reducing fuel consumption would be a critical target to achieve decarbonisation goals**. Part 1 of the NEP targets the transition of the transport system, encouraging adoption of electric vehicles and improvement of public transport.

Residential heating in Prjmnestwe constitutes the **largest component of national energy demand**, and a major source of greenhouse gas emissions (GHGs). In 2020, **heating across sectors** contributed to **51.8% of total energy consumption**, surpassing electricity generation and transport combined (National Energy Policy – Background, 2024). Within households, **heating comprised 87% of total household energy use** (space heating and water heating). The heating demand is met primarily by **gas boilers**, which leads to **high emissions and energy consumption**, hindering Prjmnestwe's net-zero goals. Part 2 of the NEP targets the electrification of domestic heating as well as establishment of mandatory building ratings to reduce the energy demand from inefficient construction.

ENERGY PROFILE

Sector	Key Characteristics	Emissions Profile / Trends
Heating	- 51.8% of total national energy use (2020) - 87% of household energy for space/water heating - 80% of residential heating from natural gas - Renewables: mainly biomass, limited heat pumps	- Largest single source of national energy-related CO₂ emissions - High emissions due to gas reliance and low renewable heat share
Electricity	- 2020 consumption: 27,979 GWh (↓ from 35,802 GWh in 2005) - 61.8% from renewables (onshore wind 61% of this) - Nuclear: 25.7% - Fossil fuels: 10.9% (↓ from 47.2% in 2005)	- Electricity sector emissions have fallen sharply since 2005 - Renewables and nuclear have displaced most coal/gas-fired generation
Transport	- 25% of total energy use - Petrol/diesel dominate - Cars/vans main mode - Freight energy demand rising - Limited electrification	- Transport is a major and persistent source of emissions - Slow progress on decarbonisation due to fossil fuel dominance
Industry	- 67% gas-dependent - 20% coal, 13% oil (esp. steel, chemicals) - Second largest end-use sector	- Industrial emissions remain high due to fossil fuel reliance - Some efficiency gains, but decarbonisation lags behind power sector
Overall Trends	- 21.4% total energy use reduction (2005–2020) - Rising energy productivity	- Overall emissions declining, driven by electricity sector - Further progress depends on heating, transport, and industry transition

Table 1. Energy Mix as per National Data with indicator of Emissions profile (Author, 2025)

POLICY MEASURES AND STRATEGIES

Part 1 – Reshaping National Mobility

Policy 1 – Electric Vehicles

Personal EVs Policy on reducing carbon emissions by Norway

Norway has enforced a policy since 1990 to promote usage of electric vehicles and finally result in a net zero carbon emission goal for newly sale cars by 2025 (Norsk elbilforening, 2024). This policy provides all kinds of advantages for the public to use electric vehicles, including tax free on purchasing electric vehicles, free parking spaces, free for toll roads, and free to use bus lane when driving (Norsk elbilforening, 2024). The policy has successfully transferred the purchasing option for family in Norway because of good welfare for vehicles purchase and after. A case study on decarbonization plan from Hinnøya in Norway suggests that promoting electrification has successfully reduced carbon emissions overall (Zhou, 2022). To achieve zero carbon emissions in the future, it is necessary to replace fuel energy system vehicles by electric vehicles. However, with several welfare options for EV purchases, the dominant vehicles in Norway are of combustion type, contributing to carbon emissions (Norsk Industri, 2022). This is due to disadvantages in EV usage, including short range and poor performance under cold weather conditions, inadequate charging stations and reduced cost-effectiveness of batteries (Gelmanova, 2018).

Evidence based energy policies and actions

With several difficulties in transferring procedure from combustion vehicles to electric vehicles, it is very difficult to achieve zero carbon emissions transferring in 2050.

Phase 1

To promote EVs, it is necessary to give the public several welfares on purchasing and using EVs like Norway. Norway has set up the welfare since 1990, but the number of road EVs starts increasing from 2014 based on the graph (Gelmanova, 2018). Therefore, Prjmnestwe needs a better welfare system to accelerate the transformation. Exemption from 30% VAT on purchase and a trade-in policy from old combustion vehicles to new electric vehicles during 2025 to 2030 are highly recommended. A trade in policy will significantly increase the willingness on purchase to consumers (Gelmanova, 2018). All toll roads and parking fees are free within the 5 years period. Since the duration is short, exemption from VAT will not be useful because it would even increase the price of products in short terms (Kucerova, 2017).

Phase 2

From 2030 to 2040, exemption from VAT will dominantly affect the willingness on purchase and Prjmnestwe should increase exemption percentage to 50% to

increase the purchase on EVs and make public get used to EVs. By 2040, sales of new combustion vehicles must be completely removed from the market, which should be the same period as Norway in 2025 (Norsk elbilforening, 2024).

Phase 3

To achieve over 90% reduction on road combustion vehicles target on private vehicles in 2050. Electric charging network, especially fast charging options, should be well set up for long distance drive for the public to feel that EVs are more convenient than combustion vehicles in order to make a full transformation. As increasing number of EVs, centralized charging may lead to grid load over peaks, voltage instability, and power factor degradation, with significant impacts especially at the distribution network level (Ashfaq, 2021). Improvement in technology is necessary to meet the requirement for the network including improving grid capacity, grid modernization, and user behavior studies for charging (Ashfaq, 2021).

Policy 2 – Public Transport

To facilitate the target of net zero by 2050, Prjmnestwe has already recognised the necessary step of shifting away from private transport reliance (according to the background document). A redesign of the current public transport system is required to increase access to reliable and expedient transit while addressing the energy trilemma.

Across Prjmnestwe, personal vehicle dominance is extremely energy-intensive accounting 61.68% of total transport energy (background document). The challenge for Prjmnestwe is to balance the needs of an expansive rural population with the needs of the urban residents.

Rural Networked Design – A Pulsing Hubs Approach

Prjmnestwe has a low population density, with a significant proportion of dwellings existing in rural areas, around 17% (background document). As the population stabilises, or decreases, it becomes imperative to connect cities to rural and exurban regions to preserve the flow of goods and services. This falls under the energy equity category of the energy trilemma, ensuring the more vulnerable regions of Prjmnestwe can access resources (transport, goods and services) equitably.

Rural public transit systems are often characterised by low information quality, limited operating hours and indirect routes which limits their effectiveness (International Transport Forum, 2021). Daniels & Mulley (2012) identify the central principles of effective public transport to be “coverage, frequency, legibility and directness”, and so the construction of Prjmnestwe’s transit system should exhibit these qualities in both urban and rural areas.

Prjmnestwe’s topology will be modelled using Petersen’s (2016) analysis of Switzerland’s rural transport network. Prjmnestwe has been described as possessing “the Highlands in the north, the Central Lowlands and the Southern Mountains” (background document) which resembles several key regions within Switzerland’s borders. There are key differences in population, GDP, cultural and

political institutions – however in terms of strictly the model for topological analysis, Prjmnestwe and Switzerland can be considered analogous.

The design Petersen (2016) proposed involved a 3-pronged strategy.

1. Train Network Expansion, utilising a hub-spoke design to boost the coverage of Prjmnestwe’s rural network coverage
2. A “pulsing timetable” in which each train station will be serviced at regular intervals (e.g. hourly), with extra trains slotted in between as needed.
3. Finally, multimodal connections that disperse out from regional centres, mirroring the structures of the train network on a smaller scale.

Spoke-Hub design will be familiar to many public transit designers as they are the predominant way in which we think of transit design today (Huang et al., 2018). The general design of this topology is that services collect in regional hubs that channel into dominate cities, much like spokes on a wheel, which can be repeated at several differing scales to make a cohesive inter-model public transit network (see Figure 2).

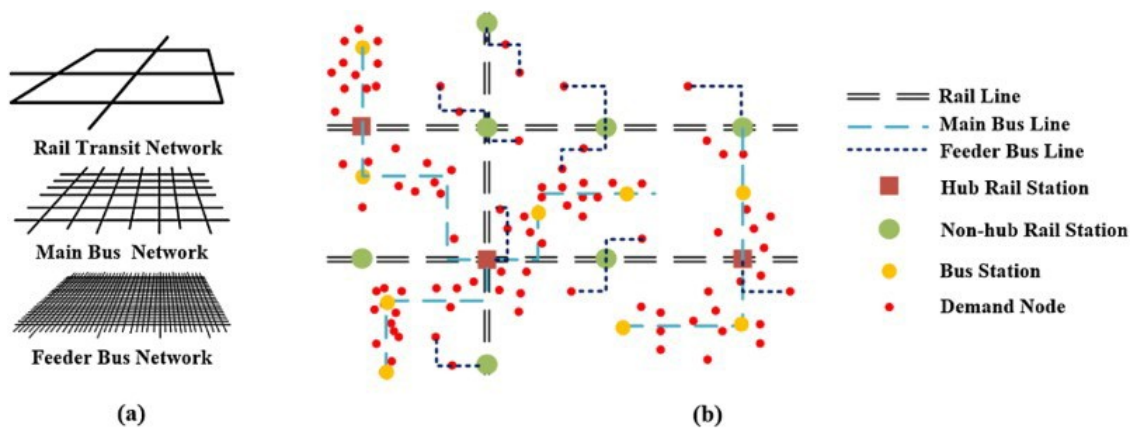


Figure 1. Multi-scalar, multimodal, spoke-hub public transit network diagram – inspiration for the design of public transport within Prjmnestwe (Huang et al., 2018).

With Prjmnestwe’s capital being in the centre of the country (according to the background document), this provides a clear, centralised hub point to begin. Radiating outwards from the capital will be newly constructed trainlines targeting regional hubs where critical services are located. Shopping centres, hospitals, schools and business centres will be the dominant indicators used to identify regional hubs (International Transport Forum, 2021). Extensive surveying should be undertaken to identify key regional hubs and lines required to limit coverage gaps.

The key organising principle on which this new rail network will be constructed is the concept of the pulse timetable, which Petersen (2016) identifies as the key to Swiss transit success.

“The Weinland (has) multiple small villages interspersed through 175 square kilometres of otherwise rural land. It has a regional population density of 1.6 persons per hectare (“p/ha”). Although the region might be expected to be completely car dependent, its public transport is able to support a journey to work mode share of 21.7%” (Petersen, 2016, p.179).

Pulse timetables are an organising principle in which transit networks arrive, leave a small window for passengers to change services and leave hubs on regular intervals so that all forms of transport possess high levels of frequency and legibility. Adopting this style of transportation is key to the new rail network of Prjmnestwe. For the rural rail network to be considered a success – we recommend setting a goal of exceeding the 21.7% journey rate achieved by the Weinland.

Now we have a rail system that is extremely efficient at bringing rural residents to and from central hubs, which is where part 3 of the transit policy is leveraged. A multi-model, local network system designed to link hubs to remote communities. Busses will be the key mode of transport for intra-communal transport, servicing the needs of residents on a similarly regular, hourly interval (adjusted as necessary to provide maximum frequency). Repeating styles of transportation on different scales has proven to be an effective form of design due to the increased coverage and legibility of its design (Daniels & Mulley, 2012). Utilising electric powered bus vehicles is viable within urban settings, but in rural spaces will be more limited. We recommend using petrol vehicles during the pilot programs and transitioning to an electric bus fleet if the program is successful. Thus Prjmnestwe will adopt this strategy as the final part of its new public transport system driving down the emissions of transport drastically.

Activity of development	Target Year
Establishment of regional centre profiles	2028
Construction on rail network begins	2033
Rail Network operational – Simultaneous to development and implementation of bus fleet	2039
Rail occupancy reaching goal of 21.7% (or above)	2043

*Timeline adapted from Johnson et al. (2013).

Table 2. Development timeline for rural rail network

Urban Networked Design – Considerable Options

Sydney’s model for urban public transit design is appropriate for Prjmnestwe, due to similarity in urban fabric. In 2021, Greater Sydney had 2,076,284 private dwellings (Australian Bureau of Statistics, 2021). Prjmnestwe has approximately 2,497,000 private dwellings, with 2,072,510 considered “urban dwellings”.

The key understanding of urban design for Sydney’s public transport networks is explained by the NSW government’s Principle 6 of Movement and Place: Plan public transport infrastructure, services and technology concurrently. Ultimately, a city’s fabric is changing continuously, therefore, there can be no perfect policy structure to perfectly match the needs of all residents. What is important for an urban public transit system is that it is adaptable and evolves over time (Government of New South Wales, 2022).

Over the next 5 years, each precinct's transport needs will be identified through surveying of residents and commuters who use the streets. Leveraging local

governments and insights of key actors will identify concerns and areas of improvement.

Simultaneously, pilot bus, tram and train expansion programs will begin to integrate peri-urban and rural connections, working from the outside in. By focusing on drivers who commute the farthest, carbon emissions are more effectively reduced. As systems become more effective, inner-city transit can scale up to fully satisfy public transit demand.

Part 2- Energy Efficient Household Heating

BACKGROUND

The reliance on natural gas in Prjmnestwe mirrors trends in the UK, where **85% of homes use gas boilers** and **residential fossil combustion contributes over 20% of emissions** (Ambrose, 2023). Despite government efforts to promote heat pumps and electric heating, gas remains dominant. 2019 figures show 52% of homes are in disrepair, 69% fail national Housing Quality Standards, and the majority are under-insulated, with only 30% fulfilling loft insulation criteria (p. 3-4). An average energy efficiency rating of EPC Band D is seen across the housing sector, indicating high heating expenses for residents and widespread low energy performance.

These inefficiencies contribute to fuel poverty, affecting 24.6% of households in 2019 (p.3). Climate conditions point to a cool, temperate climate with high humidity and winds, which further increase the heating demand (p.5). Household growth and more single-person homes will raise energy use in the absence of improvement measures.

ENERGY MIX FOR DOMESTIC HEATING

Heating Fuel	Energy Share	Relative Emissions
Natural Gas	80% of overall heat	Highest
Biomass and Biomethane	84% of renewable heat	Moderate (lower than fossil fuels, but with land-use and trade-offs; closed carbon cycle)
Heat Pumps	8% of renewable heat	Lowest

Table 3. Energy mix for domestic heating (IEA, 2022b)

The remaining share of renewable heat is done from waste heat recovery (zero emissions).

Evidence for Energy Efficiency measures- Germany’s Building Energy Act (GEG)

Germany's Building Energy Act (GEG) provides a successful example of how mandatory energy performance standards can drive significant energy savings in the building sector. Prjmnestwe can adapt this model by tailoring the standards to its climate, providing financial support for homeowners (to be detailed by Building

Council by 2026). According to the German Federal Ministry for Economic Affairs and Energy, buildings built under these standards reduce energy use by over 40% compared to traditional buildings and increase market value and comfort (DIN Standards, 2025). Implementing similar measures in Prjmnestwe is projected to reduce building energy consumption by at least 30% and lower household energy bills by eliminating fuel expenditure.

Supported by KfW Bank’s financial incentives for heating system transitions, the policy has led to widespread adoption of high-efficiency standards of energy-efficient heating (DIN Standards, 2025).

DECARBONISATION TARGETS

A phased approach ensures housing stock transitions progressively while allowing time for market adaptation (IEA, 2022a). To achieve significant decarbonization in the heating sector, this policy establishes the following targets, aligned with Prjmnestwe's commitment to net-zero emissions by 2050. These targets address both short-term improvements and the long-term transformation of the heating system.

Timeframe	Key Targets
Short-Term (By 2025) Effective Immediately	- Enforce energy efficiency building code drafts for new constructions starting in 2026 - Retrofit 15% of existing homes with insulation and heating upgrades, prioritising fuel-poor households - Require low-carbon heating systems in all new residential buildings - Train AEC workers in retrofit and heat pump installations - Launch awareness campaigns with a focus on Northern Prjmnestwe
Medium-Term (By 2030)	- Upgrade 1.5 million homes (~60% of older stock) to minimum EPC Band C - Achieve 30% market share for low-carbon heating in existing homes - Ban new gas boiler installations in new builds and major renovations - Connect 200,000 homes to renewable district heating - Reduce fuel poverty by 75% from 2019 levels - Mandate rental properties meet minimum efficiency standards (no below Band D by 2028, Band C by 2030)
Long-Term (By 2040)	- Phase out all fossil-fuel heating systems in residential buildings - Ensure >90% of homes use electric or renewable heating - Achieve universal compliance with modern insulation standards - Cut residential emissions by 90% from 2020 - Make all new renovations net-zero ready

Table 4. Streamlined Measures for Heating Demand Management

PROPOSALS

Section 1- Establishment of Energy Efficiency Standards

Prjmnestwe's houses face problems with climate change adaptation measures and high energy bills, owing to inefficient heating and insulation measures. According to IEA (2022a), establishment of energy efficiency mandates such as building codes lead to a reduction in energy bills. These can be effectively implemented through mandatory code compliance and disclosure, detailed further, without which building approvals may be declined or fined.

1.1. Deep Retrofits and Insulation Upgrades

Improving the thermal efficiency of existing homes is the most cost-effective and immediate strategy for reducing heating demand and emissions (Ürge-Vorsatz et al., 2020). Retrofitting measures include thermal insulation and air gap sealing to prevent heat transfer shown in Figure 2. Energy retrofits **can reduce household heating energy consumption and associated CO₂ emissions by 30-50% or more** (Broer et al., 2021; Rosenow et al., 2020) in older housing stock with low baseline efficiency.

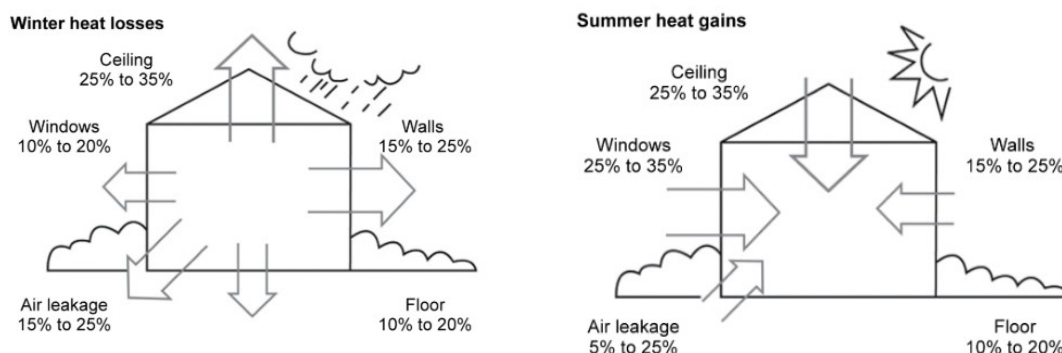


Figure 2. Heat losses and gains in the absence of effective insulation and airtightness measures (Mosher & McGee, 2013)

Targeted government programs for retrofit upgrades are effective in accelerating uptake when focused on low-income or fuel-poor households (Galvin & Sunikka-Blank, 2012). Workforce training in retrofit techniques is also recommended (Killip et al., 2020).

1.2. Strengthening Building Codes

Stringent building energy codes and minimum energy performance standards are critical for locking in long-term efficiency gains and preventing “carbon lock-in” in new construction (Haar, 2020; The World Bank, 2024). Mandating **high insulation** and **airtightness** is essential to progress to an efficient, low-emission housing stock (IEA, 2022a). **Loft Insulation of 300mm (Background Document) will be**

mandatory in all new constructions effective 2026, along with sealings and maximum air change rate of 3 h⁻¹ (DIN Standard).

Introducing phased requirements for existing buildings to reach minimum energy performance benchmarks (e.g., EPC Band C) can drive market transformation (Kerr et al., 2017), especially when combined with financial and technical support for homeowners and landlords (Killip et al., 2020).

1.3. Energy Performance Certificate (EPC)

A	B	C	D	E	F	G
Net Zero	Energy Efficient	Baseline	Below Baseline- Allowable only for Existing Stock starting 2026			

Figure 3. Rating System Mandates for New and Existing Constructions (Author, 2025)

- **EPCv2- New Buildings Standard:** New buildings must achieve **a minimum of Band B by 2030**, progressing to **Band A by 2035**, to meet the decarbonisation target.
- **EPCv2- Existing Building Standard:** The Existing Housing Stock will follow a weighted rating system. All buildings must be retrofitted to reach **band C by 2030**, progressing to **band A by 2040** as per the decarbonisation target.
- **Band C** will represent the **Minimum Acceptable Performance** from **January 2028**.
- **Life Cycle Assessments (LCA)** will be conducted (Scope A1-A5)

1.3.1. Reference Building Approach for Rating Assessments

Energy Performance Rating will be calculated by comparing the proposed building’s model to a reference case, which represents the same geometry and activity but with the minimum acceptable performance.

Phase	Period	Minimum Energy Reduction Target for Code Compliance	Alignment with Decarbonisation Goals
1	2026-2028	5-10%	Consistent with early retrofit efforts and building code tightening recommended by SBTi (2024) and IEA (2023). Enables market adaptation and workforce training.
2	2028-2030	20-30%	Reflects accelerated retrofit and efficiency improvements needed to halve building emissions by 2030 (IEA, 2023). Matches UNEP (2025) call for zero-carbon building codes by 2028.
3	2030-2035	50-60%	Supports deep retrofit and near-net-zero building standards for new and existing buildings, consistent with EU EPBD pathway for zero emissions by 2030-35 (European Parliament, 2023).
4	2035-	80%+	Aligns with near-full decarbonisation of building operations and

	2040		phase-out of fossil fuel heating by 2040, as recommended by SBTi (2024) and IEA (2023) pathways
--	------	--	---

Table 5. Minimum requirement for EPC accreditation aligned with global recommendations (Author, 2025)

1.4. Implementation Measures

1) Mandatory Building Code and EPC Compliance

Minimum standards for energy-efficient buildings will be enforced in January 2026, for different classifications of buildings, pilot phase focused on single and multi-unit residential dwellings. All buildings must be rated for efficiency by an **Accredited House Energy Assessor** and demonstrate code requirements.

2) Mandatory Energy Rating Disclosure

Every home must display a valid energy rating certificate. This transparency will influence buyer and renter choices due to low operational and maintenance costs.

3) Government Incentives

- Need-based aid** for retrofit measures in **low-income households**.
- Utility Bill Credits** for all buildings rated at Band A starting in 2026
- Allocation of 20% of Carbon Tax** collected from **businesses failing to meet emission targets** to support **low-emission initiatives**.

Section 2 - Transition to Electricity-based Heating

Transition from fossil-fuel-based heating to renewable-powered systems is vital for decarbonising domestic heating (IEA, 2022a). Heat pumps, which currently provide 8% of the heating needs, are highly efficient, producing **3-5 units of heat for every unit of electricity consumed** (Staffell et al., 2022). Life-cycle assessments show that replacing gas boilers with heat pumps can reduce household heating emissions by 70-90%, on decarbonised grids (Fawcett et al., 2019; Love et al., 2017).

Direct financial incentives for heat pump installation, **regulatory barrier reduction** and **investment in grid infrastructure** to manage rising energy consumption will all promote widespread adoption (Lowes et al., 2020). In densely populated areas, renewable-powered district heating networks utilising large-scale heat pumps, geothermal, or waste heat can improve efficiency and cut emissions at scale (Lund et al. 2018).

Many countries have set deadlines after which **fossil-fuel heating systems are banned in new buildings**- an approach associated with significant emissions reductions (Rosenow et al., 2020; The World Bank, 2024).

Section 3- Public Awareness and Behavioural Change

Public awareness campaigns inform residents about the benefits of efficient heating and available support programs, while also promoting simple behavioural changes, such as optimising thermostat settings or draft-proofing, can yield immediate energy savings (Abrahamse et al., 2005). Research suggests that well-designed information and engagement programs can reduce household energy consumption by 10–20% (Abrahamse et al., 2005; Wilson et al., 2015). Training for installers and builders is important to ensure the supply chain is prepared for new technologies and can provide reliable advice to households (Killip et al., 2020). The campaign aims to target the northern population to address ongoing energy costs.

Policy Implementation Framework

NEP 2025 Implementation Timeline	2026	2027	2028	2029	2030	2032	2035	2040	2045	2050
Short-Term: Foundation (2025–2030)										
Regulatory Frameworks, Incentives, & Infrastructure										
Launch EV & Retrofit Incentives										
Develop Public EV Charging & Pilot Public Transport										
Updated Building Codes & Retrofit Programs										
Awareness & Training Programs										
Medium-Term: Acceleration (2030–2040)										
Expand EV Charging & Public Transport										
Phase out Gas Boilers in Homes										
Upgradation for Existing Buildings to Band B minimum										
District Heating Networks										
Stricter Vehicle Emission Standards										
Long-Term Decarbonization (2040–2050)										
Zero Emissions Heating										
Highest Insulation Standards (Passivhaus eq.)										
Net-Zero Ready New & Renovated Homes										
Near-Zero Emissions in Heating & Transport										

Table 6. Gantt Chart for NEP 2025, Implementation timelines and expected achievement (Author, 2025)

Performance Indicator Metrics

- Increased share of EVs in new car sales.
- Growth in public transport ridership.
- Reduction in energy consumption for residential heating.
- Increase in the number of homes above band C.
- Emissions Evaluation 2030- Reductions in Heating, Building-related and Vehicular energy use

Conclusion

The energy policy is a directed approach designed to target the biggest contributors to Prjmnestwe's greenhouse gas emissions. By tackling the issues of housing and transport directly, the government curbs the emissions generated by energy-intensive activities while simultaneously improving the lives of its citizens.

Within transport, EV uptake has proven to be a cornerstone of environmental action. Due to Prjmnestwe's reliance on private transport, there is limited scope for rapid decarbonisation. While the expansive rail network is being constructed, EVs provide market options for greater sustainable movement without disrupting the lifestyles of those who rely on their motor vehicles. Once the rail network is fully operational, the opportunity to move around Prjmnestwe without a car becomes viable. Rural communities become deeply connected to regional centres, and the island transforms into a more mobile and interconnected society.

Housing sector transformation is equally critical. Residential heating remains the largest source of energy demand and a major contributor to national emissions. The policy outlines a phased and inclusive pathway to decarbonise household heating by mandating energy performance standards, promoting retrofits, and transitioning to electric and renewable heating systems. Measures such as EPC rating systems, heat pump subsidies, and public awareness campaigns are designed to reduce fuel poverty, cut emissions, and improve living standards—particularly for vulnerable populations. These housing policies not only reduce energy bills but also enhance thermal comfort and resilience in the face of climate variability.

Ultimately, these policies are a significant step towards energy efficiency and its transition away from fossil fuels, and it is important to accommodate futuristic considerations. The current actions will have a profound impact on national industry and the country's positioning in international trade. As the energy demand is set to rise, a reliable and sustainable supply of energy becomes essential. Given Prjmnestwe's climatic limitations—low solar irradiation and seasonal wind inconsistency—further investment in nuclear baseload power is necessary to stabilise the grid. We recommend that the government prevent the decline of its nuclear sector and initiate a policy report to explore future expansion in this area.

References

- ABS. (2021). *2021 Greater Sydney, Census all persons QuickStats | Australian Bureau of Statistics*. www.abs.gov.au. <https://www.abs.gov.au/census/find-census-data/quickstats/2021/1GSYD>
- Ashfaq, M., Butt, O., Selvaraj, J., & Rahim, N. (2021). Assessment of electric vehicle charging infrastructure and its impact on the electric grid: A review. *International Journal of Green Energy*, 18(7), 657–686. <https://doi.org/10.1080/15435075.2021.1875471>
- Daniels, R., & Mulley, C. (2012). Planning public transport networks—the neglected influence of topography. *Journal of Public Transportation*, 15(4), 23–41. <https://doi.org/10.5038/2375-0901.15.4.2>
- Dimanchev, E., Davood Qorbani, & Magnus Korpås. (2022). *Electric Vehicle Adoption Dynamics on the Road to Deep Decarbonization*. 177–205. <https://doi.org/10.1002/9783527831425.ch8>
- DIN Standards. (2025). Energy efficiency in buildings. *Dinmedia.de*. <https://doi.org/1048604/aa7c99f2fca6e28b4370fd275a9bdee2>
- European Parliament. (2023). *European Parliament P9_TA(2023)0068 Energy performance of buildings (recast)*. https://www.europarl.europa.eu/doceo/document/TA-9-2023-0068_EN.pdf
- GEG. (2023). *Buildings Energy Act (GEG; consolidated 2023 version) - Climate Change Laws of the World*. Climate-Laws.org. https://climate-laws.org/documents/buildings-energy-act-geg_30e5?id=buildings-energy-act-geg_ec47
- Gelmanova, Z. S., Zhabalova, G. G., Sivyakova, G. A., Lelikova, O. N., Onishchenko, O. N., Smailova, A. A., & Kamarova, S. N. (2018). Electric cars. Advantages and disadvantages. *Journal of Physics: Conference Series*, 1015(5), 052029. <https://doi.org/10.1088/1742-6596/1015/5/052029>
- Gillott, M. C., Loveday, D. L., White, J., Wood, C. J., Chmutina, K., & Vadodaria, K. (2016). Improving the airtightness in an existing UK dwelling: The challenges, the measures and their effectiveness. *Building and Environment*, 95, 227–239. <https://doi.org/10.1016/j.buildenv.2015.08.017>
- Huang, D., Liu, Z., Fu, X., & Blythe, P. T. (2018). Multimodal transit network design in a hub-and-spoke network framework. *Transportmetrica A: Transport Science*, 14(8), 706–735. <https://doi.org/10.1080/23249935.2018.1428234>
- IEA. (2022a). *Energy Efficiency - Energy System*. IEA. <https://www.iea.org/energy-system/energy-efficiency-and-demand/energy-efficiency>
- IEA. (2022b). *Renewable heat – Renewables 2022 – Analysis*. IEA. <https://www.iea.org/reports/renewables-2022/renewable-heat>
- IEA. (2023). *The energy efficiency policy package: key catalyst for building decarbonisation and climate action – Analysis*. IEA.

<https://www.iea.org/commentaries/the-energy-efficiency-policy-package-key-catalyst-for-building-decarbonisation-and-climate-action>

ITF. (2021). Innovations for better rural mobility. In *OECD Publications*.
<https://doi.org/10.1787/6dbf832a-en>.

Johnson, D., Jackson, J., & Nash, C. (2013). The wider value of rural rail provision. *Transport Policy*, 29, 126–135. <https://doi.org/10.1016/j.tranpol.2013.04.007>

Kerr, N., Gouldson, A., & Barrett, J. (2017). The rationale for energy efficiency policy: Assessing the recognition of the multiple benefits of energy efficiency retrofit policy. *Energy Policy*, 106, 212–221.
<https://doi.org/10.1016/j.enpol.2017.03.053>

Kučerová, V. (2017). VAT and its influence on buying behaviour (Institute of Economic Research Working Papers No. 52/2017). Institute of Economic Research (IER), Toruń. <https://hdl.handle.net/10419/219875>

Mosher, M., & McGee, C. (2013). *Insulation | YourHome*. www.yourhome.gov.au; Australian Government. <https://www.yourhome.gov.au/passive-design/insulation>

Norsk elbilforening. (2024). *Norwegian EV policy*. Elbil.no.
<https://elbil.no/english/norwegian-ev-policy/>

Norsk Industri. (2022). *ENERGY TRANSITION NORWAY 2022 a National Forecast to 2050*. https://www.norskindustri.no/siteassets/dokumenter/rapporter-og-brosjyrer/energy-transition-norway/2022/energy-transition-norway-2022_web.pdf

Petersen, T. (2016). Watching the Swiss: A network approach to rural and exurban public transport. *Transport Policy*, 52, 175–185.
<https://doi.org/10.1016/j.tranpol.2016.07.012>

Place, M. and. (2022, July 14). *Principle 6: Plan public transport infrastructure, services and technology concurrently*. nsw.gov.au.
<https://www.movementandplace.nsw.gov.au/place-and-network/guides/network-planning-precincts-guide/network-planning-and-design-principles/principle-6-plan-public-transport-infrastructure-services-and-technology-concurrently>

SBTi. (2024). *Science Based Targets initiative- BUILDINGS SECTOR SCIENCE-BASED TARGET-SETTING CRITERIA- Version 1*.
<https://sciencebasedtargets.org/resources/files/SBTi-Buildings-Criteria.pdf>

Srivastava, J., & Chakravarti, D. (2011). Price Presentation Effects in Purchases Involving Trade-Ins. *Journal of Marketing Research*, 48(5), 910–919.
<https://doi.org/10.1509/jmkr.48.5.910>

Sunikka-Blank, M., & Galvin, R. (2012). Introducing the prebound effect: the gap between performance and actual energy consumption. *Building Research & Information*, 40(3), 260–273. <https://doi.org/10.1080/09613218.2012.690952>

The World Bank. (2024). *Mapping Energy Efficiency: A Global Dataset on Building Code Effectiveness and Compliance INITIAL FINDINGS AND INSIGHTS FROM THE DATA COLLECTED*.
https://www.worldbank.org/content/dam/sites/buildinggreen/doc/building_green_main_findings.pdf

UNEP. (2025). *Emissions from building sector stopped rising for the first time since 2020, UN finds*. UN Environment.
<https://www.unep.org/news-and-stories/press-release/emissions-building-sector-stopped-rising-first-time-2020-un-finds>

Zhou, W., Hagos, D. A., Stikbakke, S., Huang, L., Cheng, X., & Onstein, E. (2022). Assessment of the impacts of different policy instruments on achieving the deep decarbonization targets of island energy systems in Norway – The case of Hinnøya. *Energy*, 246, 123249. <https://doi.org/10.1016/j.energy.2022.123249>

Acknowledgements

Quillbot and Grammarly have been used to paraphrase original writing to adhere to the word limits and improve grammatical errors. Perplexity.AI was used for preliminary research but not for content generation.